

Task4

Title

Student Performance Analysis System using Classes, Objects and Arrays

Objective

To design and implement a **student performance analysis system** in Java using Object-Oriented Programming principles to calculate total marks, average, percentage, grade, highest and lowest marks, result, and overall academic performance.

System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Monitor

Software Requirements

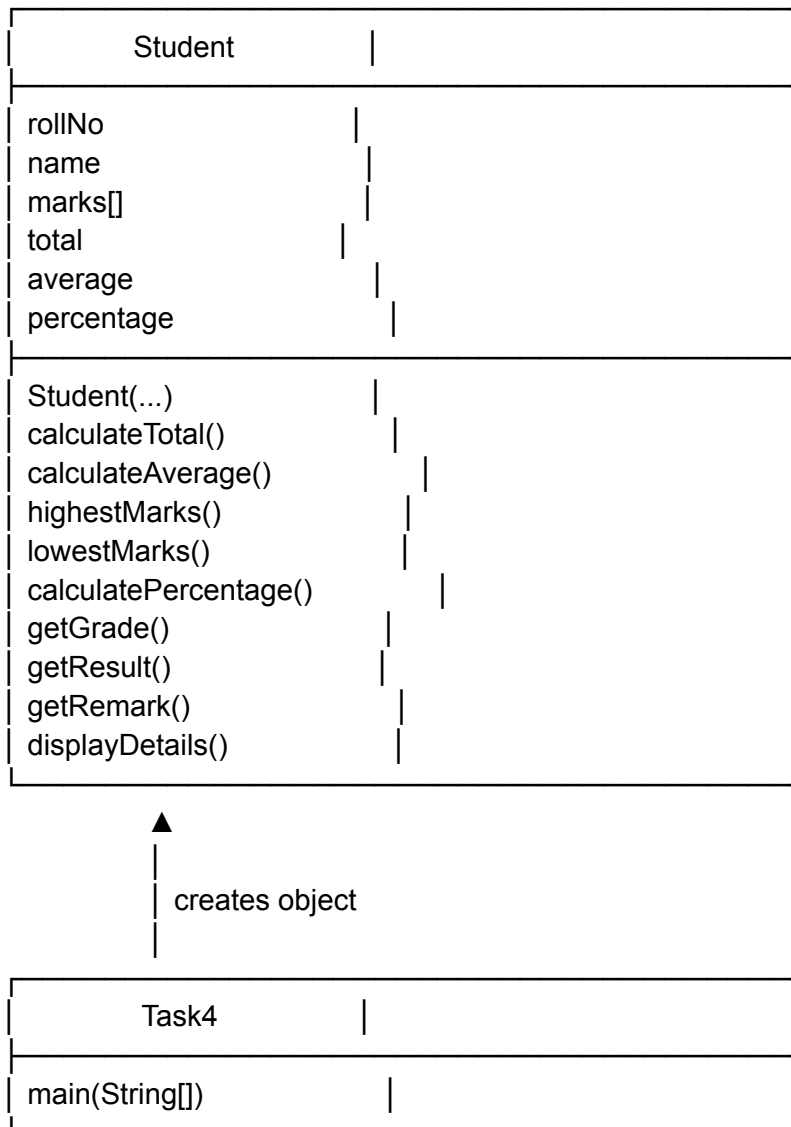
- JDK 8 or above
- VS Code / Eclipse / IntelliJ IDEA
- Command Prompt / PowerShell

Algorithm / Logic Description

1. Create a **Student** class with roll number, name, marks array, total, average and percentage.
2. Create a constructor to initialize the student details.
3. Store marks of five subjects in an integer array.
4. Use a loop to calculate the total marks.
5. Calculate the average marks.
6. Use **Math.max()** to find the highest marks.
7. Use **Math.min()** to find the lowest marks.
8. Calculate the percentage from the total marks.

9. Use `Math.round()` to round the percentage to two decimal places.
10. Use `trim()`, `toUpperCase()` and `length()` to process the student's name.
11. Assign the grade based on the percentage.
12. Determine whether the student has passed or failed.
13. Display a performance remark based on the percentage.
14. Create a `Student` object in the `main()` method.
15. Display the complete student performance report.

UML Class Diagram



Source Code

```
import java.util.Scanner;
```

```
class Student {

    int rollNo;
    String name;
    int marks[];
    int total;
    double average;
    double percentage;

    Student(int rollNo, String name, int marks[]) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }

    int calculateTotal() {
        total = 0;

        for (int i = 0; i < 5; i++) {
            total = total + marks[i];
        }

        return total;
    }

    double calculateAverage() {
        average = total / 5.0;
        return average;
    }

    int highestMarks() {
        int high = marks[0];

        for (int i = 1; i < 5; i++) {
            high = Math.max(high, marks[i]);
        }

        return high;
    }

    int lowestMarks() {
        int low = marks[0];
```

```

    for (int i = 1; i < 5; i++) {
        low = Math.min(low, marks[i]);
    }

    return low;
}

double calculatePercentage() {
    percentage = total * 100.0 / 500;
    return Math.round(percentage * 100.0) / 100.0;
}

String getGrade() {

    if (percentage >= 90) {
        return "A+";
    }
    else if (percentage >= 80) {
        return "A";
    }
    else if (percentage >= 70) {
        return "B";
    }
    else if (percentage >= 60) {
        return "C";
    }
    else if (percentage >= 50) {
        return "D";
    }
    else {
        return "F";
    }
}

String getResult() {

    if (percentage >= 50) {
        return "PASS";
    }
    else {
        return "FAIL";
    }
}

```

```

String getRemark() {

    if (percentage >= 90) {
        return "Excellent Performance";
    }
    else if (percentage >= 80) {
        return "Very Good Performance";
    }
    else if (percentage >= 70) {
        return "Good Performance";
    }
    else if (percentage >= 60) {
        return "Average Performance";
    }
    else if (percentage >= 50) {
        return "Needs Improvement";
    }
    else {
        return "Poor Performance";
    }
}

```

```

void displayDetails() {

    String formattedName = name.trim().toUpperCase();

    System.out.println("\n----- STUDENT PERFORMANCE REPORT -----");
    System.out.println("Roll Number    : " + rollNo);
    System.out.println("Student Name    : " + formattedName);
    System.out.println("Name Length     : " + formattedName.length());
    System.out.println("Total Marks     : " + calculateTotal());
    System.out.println("Average Marks   : " + calculateAverage());
    System.out.println("Highest Marks   : " + highestMarks());
    System.out.println("Lowest Marks    : " + lowestMarks());
    System.out.println("Percentage      : " + calculatePercentage() + "%");
    System.out.println("Grade           : " + getGrade());
    System.out.println("Result          : " + getResult());
    System.out.println("Remark          : " + getRemark());
}
}

```

```

public class Task4 {

    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter Roll Number: ");
int rollNo = sc.nextInt();

sc.nextLine();

System.out.print("Enter Student Name: ");
String name = sc.nextLine();

int marks[] = new int[5];

System.out.println("Enter marks of 5 subjects:");

for (int i = 0; i < 5; i++) {
    marks[i] = sc.nextInt();
}

Student s = new Student(rollNo, name, marks);

s.displayDetails();

sc.close();
}
}
```

Compilation and Execution Commands

Compile

javac Task4.java

Execute

java Task4

Output Screen

```
Enter Roll Number: 101
Enter Student Name: Naveen
Enter marks of 5 subjects:
85
90
```

78
88
92

----- STUDENT PERFORMANCE REPORT -----

Roll Number : 101
Student Name : NAVEEN
Name Length : 6
Total Marks : 433
Average Marks : 86.6
Highest Marks : 92
Lowest Marks : 78
Percentage : 86.6%
Grade : A
Result : PASS
Remark : Very Good Performance

```
92' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-  
ct\bin' 'Task4'
```

```
Enter Roll Number: 248  
Enter Student Name: naveen  
Enter marks of 5 subjects:  
90  
95  
88  
94  
92
```

----- STUDENT PERFORMANCE REPORT -----

Roll Number : 248
Student Name : NAVEEN
Name Length : 6
Total Marks : 459
Average Marks : 91.8
Highest Marks : 95
Lowest Marks : 88
Percentage : 91.8%
Grade : A+
Result : PASS
Remark : Excellent Performance